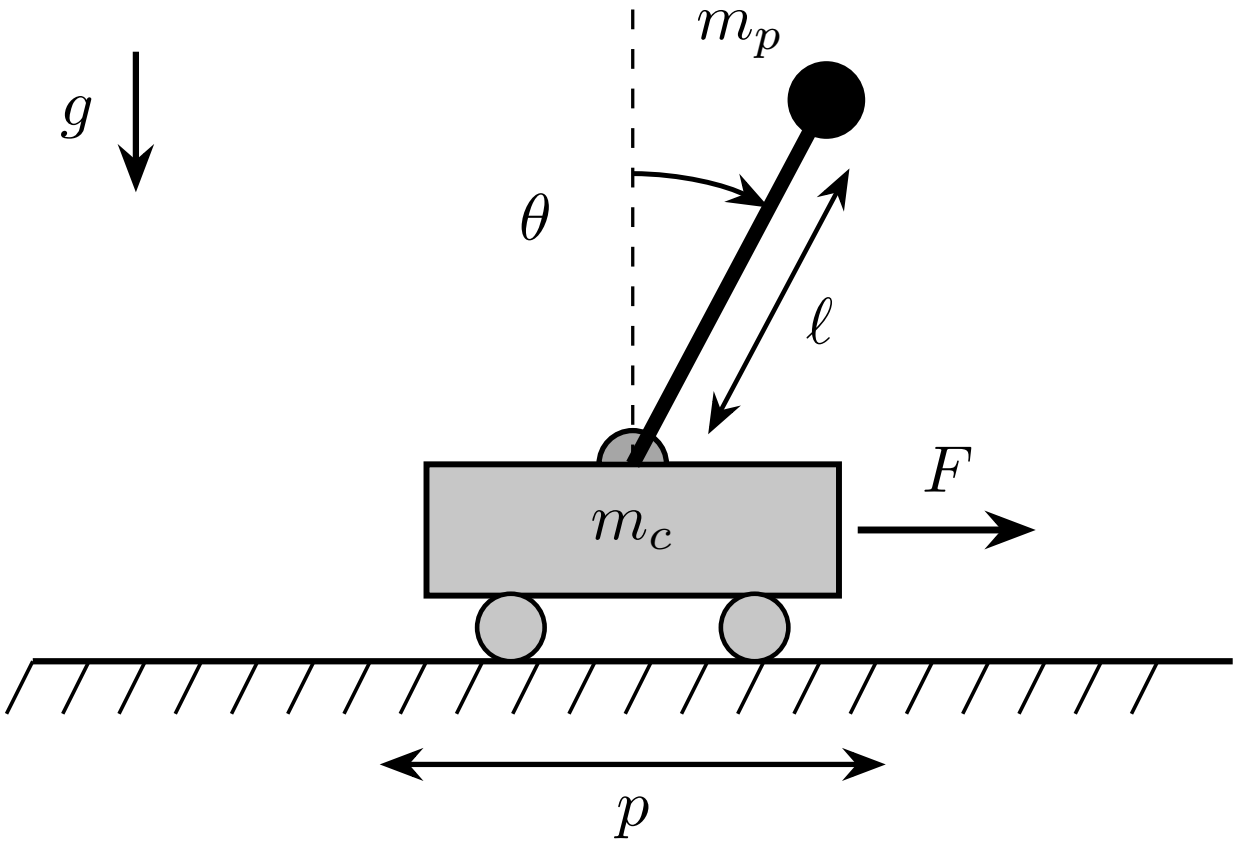

$$x = \begin{bmatrix} \theta \\ p \\ \dot{\theta} \\ \dot{p} \end{bmatrix}$$

$$u = F, \quad |F| \leq F_{\max}$$